

# Data curation strategies for genomic language models

Gonzalo Benegas, Eric Czech  
Open Athena



## Abstract

- **Pretraining data composition** is widely recognized as a key driver of LLM performance, but its role in **genomic language models** (gLMs) has not been systematically studied.
- We investigate gLM data curation along two axes: **functional regions** (e.g. CDS, promoters) and **evolutionary timescales** (humans → animals).

## gLM overview

**Training:** maximize the likelihood of sequences from reference genomes (typically from healthy individuals).

$$\text{maximize } \log P(\dots \text{CACTTGGAT} \dots)$$

**Zero-shot variant effect prediction (VEP):** score a variant by how much it changes the likelihood compared to the reference. Low likelihood → likely deleterious.

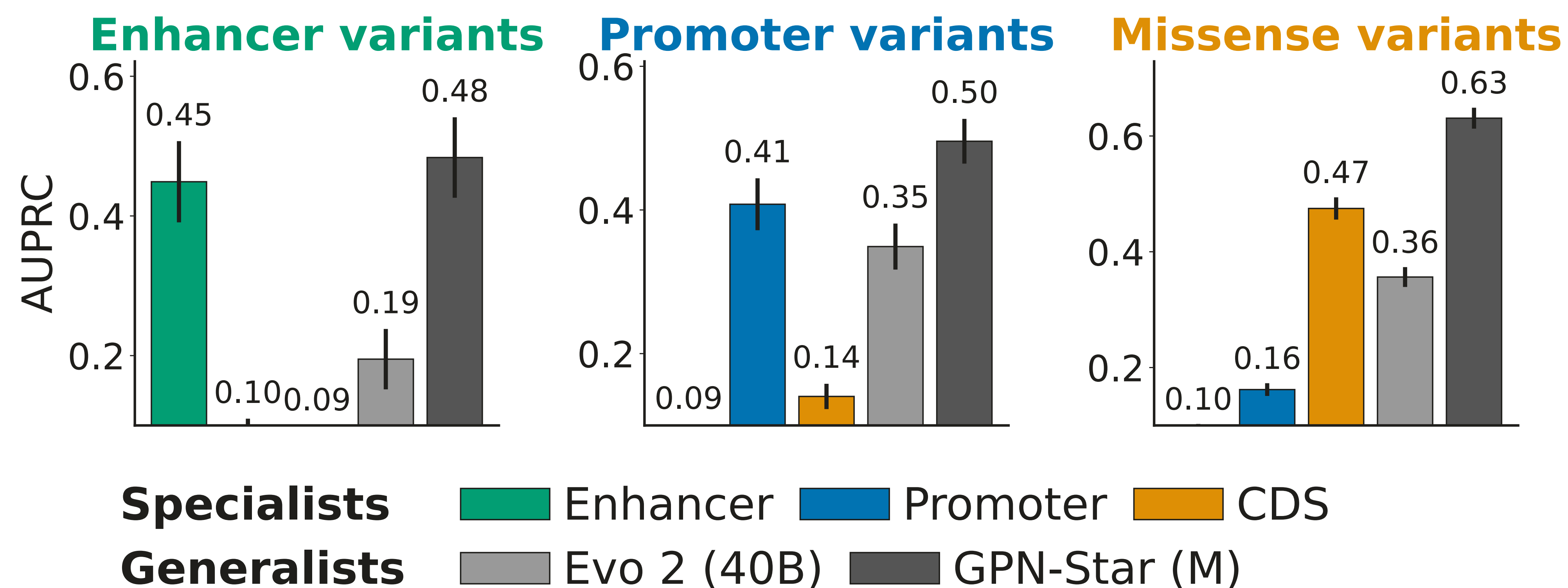
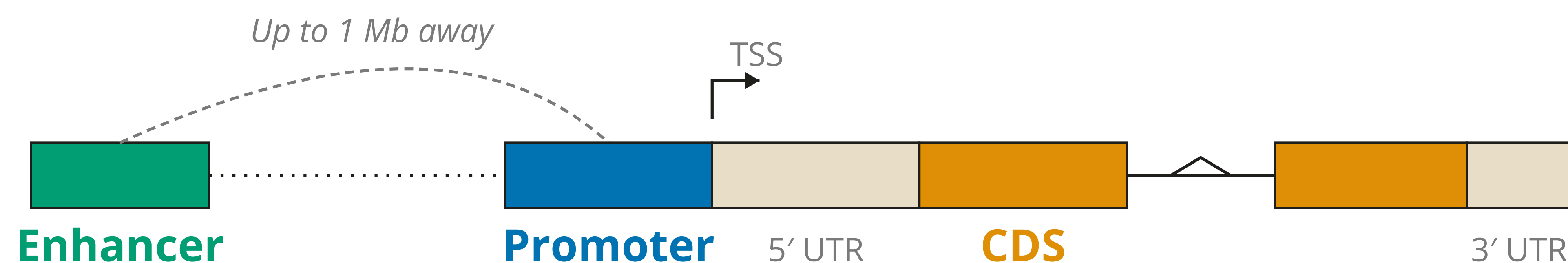
$$\text{LLR} = \log \frac{P(\dots \text{CACT} \text{CGGAT} \dots)}{P(\dots \text{CACTTGGAT} \dots)}$$

## Methods

- Standard architecture and training objective: **Qwen3 autoregressive Transformer**.
  - Reuse LLM infrastructure and modeling science; focus on data.
- **Alignment-free:** applicable to any species without whole-genome alignments.
- **Context size: 256 bp.**
  - Focus on individual functional elements (e.g. exons, enhancers) and iterate faster.
- **Model size: ~1B** (currently exploring scaling).
- Data sources: **RefSeq annotation** for genic regions, **ENCODE SCREEN** + sequence alignment for enhancers.
- Human VEP evaluation: classify **Mendelian pathogenic vs. gnomAD high-frequency** variants (similar to TraitGym).

## One region at a time

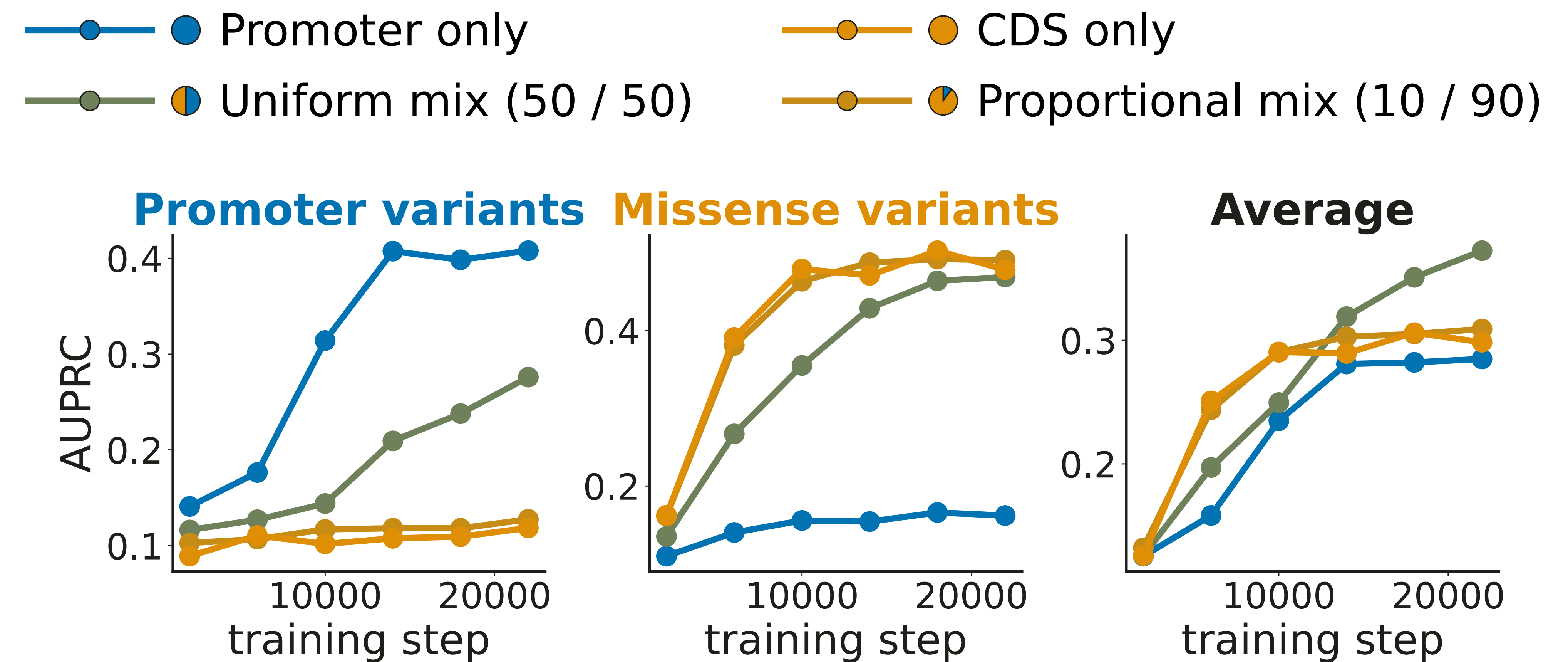
- The genome is very heterogeneous; coding sequences (CDS) and regulatory regions have different grammars.
- We first trained specialist models, each on a single region of the genome (initial timescale: animals for CDS/promoters, mammals for enhancers).



- Each specialist achieves good performance in VEP tasks on its trained region.
- Specialists often outperform Evo 2 (a much larger model), but not the alignment-based GPN-Star.

## How to mix regions?

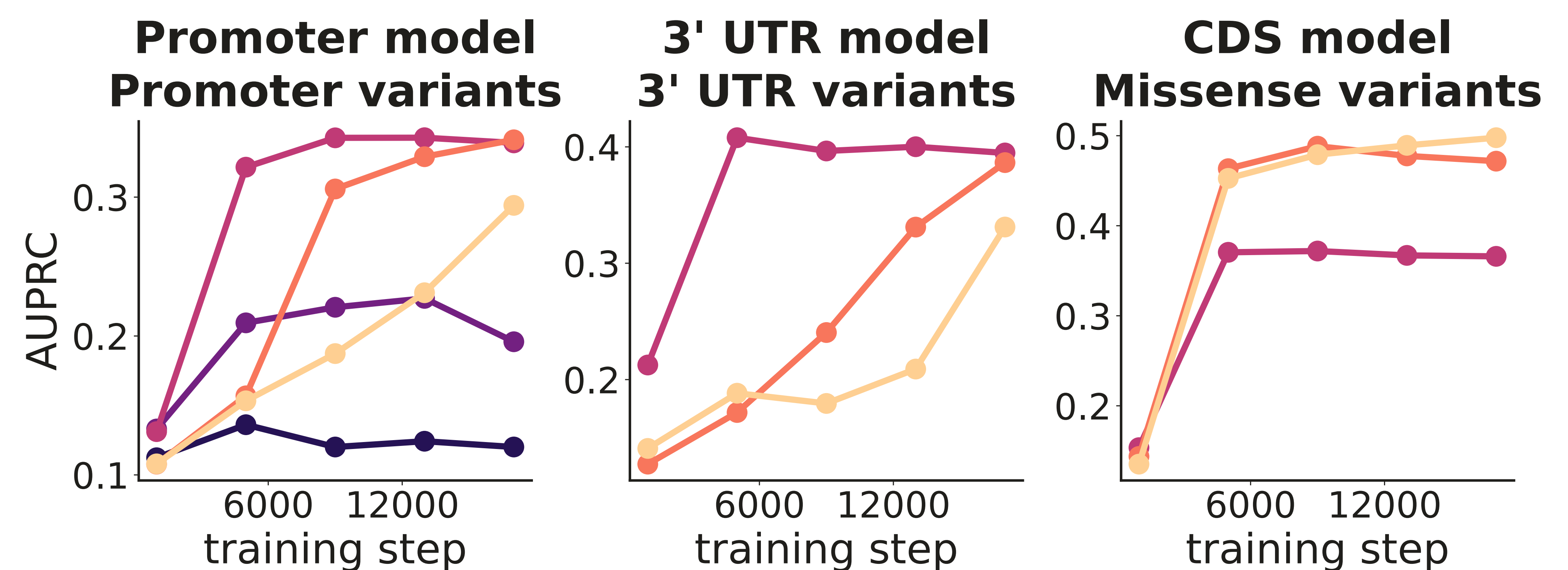
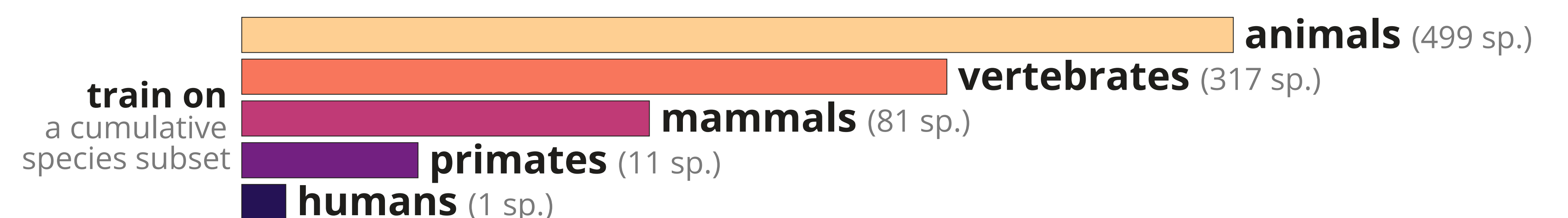
- How can we build a generalist model? Train on the union of the different regions?
- We evaluated this common approach (proportional mixing) against balanced sampling (uniform mixing).



- Proportional mixing shows poor performance on the minority region (promoters).
- Uniform mixing allows good progress across both promoter and missense variants.

## Which evolutionary timescale?

- If we have a target species of interest (e.g. human), which species should we train on?
- There is a tradeoff between dataset size, diversity, and evolutionary relevance.



- For regulatory regions (promoters, 3' UTR), the mammals model converges fast; vertebrates / animals climb more slowly but may surpass it with more compute.
- In the more conserved CDS region, the value of larger evolutionary timescales is much more evident.

## Summary and outlook

- We explore data curation strategies along two separate axes: functional regions and evolutionary timescales.
- **Balanced sampling across regions shows robust improvements.** The **optimal evolutionary timescale is less clear** and warrants further investigation.
- As next steps, we are exploring how data curation recipes interact with varying **model scale**. We are also interested in transfer learning applications such as **gene expression prediction**.
- Inspired by the Marin project, we go beyond public code/model/data: **the research process itself is open**, with all experiments preregistered and publicly available from day 1.